## a Boson: strict-cone screening; group-only symmetry fallback

columns: $J_2^T g J_2$, $J_2^2$, and $J_1, J_2$ commuting (comm.) or anticommuting (anti.)

rows: $J_1^T g J_1$, $J_1^2$

| | g / I / comm. | g / I / anti. | g / -I / comm. | g / -I / anti. | -g / I / comm. | -g / I / anti. | -g / -I / comm. | -g / -I / anti. |
|---|---|---|---|---|---|---|---|---|
| g, I | B01 d=4 group only no theorem from original $J_1, J_2$ | B02 d=4 strict matrix cone $\eta = iJ_2 g$ fixed-$N$ weight | B03 d=2 group only no theorem from original $J_1, J_2$ | B04 d=4 group only $\Theta(+,-)$: $W_B = \lvert w \rvert^2$ | B05 d=2 strict cone $\eta = J_2 g$ semigroup branch | B06 d=4 group only no theorem from original $J_1, J_2$ | B07 d=4 group only $\Theta(+,-)$: $W_B = \lvert w \rvert^2$ | B08 d=4 group only no theorem from original $J_1, J_2$ |
| g, -I | B09 d=2 group only no theorem from original $J_1, J_2$ | B10 d=4 strict matrix cone $\eta = iJ_2 g$ fixed-$N$ weight | B11 d=4 group only no theorem from original $J_1, J_2$ | B12 d=4 group only $\Theta(-,-)$: no positive particle space | B13 d=4 strict cone $\eta = J_2 g$ semigroup branch | B14 d=2 group only no theorem from original $J_1, J_2$ | B15 d=4 group only $\Theta(-,-)$: Kramers physical at d=4 | B16 d=4 group only no theorem from original $J_1, J_2$ |
| -g, I | B17 d=2 strict matrix cone $\eta = iJ_2 g$ fixed-$N$ weight | B18 d=4 group only no theorem from original $J_1, J_2$ | B19 d=4 group only $\Theta(+,-)$: $W_B = \lvert w \rvert^2$ | B20 d=2 group only no theorem from original $J_1, J_2$ | B21 d=4 group only no theorem from original $J_1, J_2$ | B22 d=2 strict cone $\eta = J_2 g$ semigroup branch | B23 d=4 group only no theorem from original $J_1, J_2$ | B24 d=4 group only $\Theta(+,-)$: $W_B = \lvert w \rvert^2$ |
| -g, -I | B25 d=4 strict matrix cone $\eta = iJ_2 g$ fixed-$N$ weight | B26 d=4 group only no theorem from original $J_1, J_2$ | B27 d=4 group only $\Theta(-,-)$: Kramers physical at d=4 | B28 d=4 group only no theorem from original $J_1, J_2$ | B29 d=4 group only no theorem from original $J_1, J_2$ | B30 d=4 strict cone $\eta = J_2 g$ semigroup branch | B31 d=8 group only no theorem from original $J_1, J_2$ | B32 d=4 group only $\Theta(-,-)$: no positive particle space |

## b Fermion: strict-cone screening; group-only symmetry fallback

columns: $J_2^T g J_2$, $J_2^2$, and $J_1, J_2$ commuting (comm.) or anticommuting (anti.)

rows: $J_1^T g J_1$, $J_1^2$

| | g / I / comm. | g / I / anti. | g / -I / comm. | g / -I / anti. | -g / I / comm. | -g / I / anti. | -g / -I / comm. | -g / -I / anti. |
|---|---|---|---|---|---|---|---|---|
| g, I | F01 d=2 group only no theorem from original $J_1, J_2$ | F02 d=4 group only no theorem from original $J_1, J_2$ | F03 d=4 group only no theorem from original $J_1, J_2$ | F04 d=4 strict cone $\eta = iJ_2 g$ semigroup branch | F05 d=2 group only no theorem from original $J_1, J_2$ | F06 d=2 group only no theorem from original $J_1, J_2$ | F07 d=2 strict cone $\eta = J_2 g$ semigroup branch | F08 d=2 group only no theorem from original $J_1, J_2$ |
| g, -I | F09 d=4 group only no theorem from original $J_1, J_2$ | F10 d=4 group only $\Theta(+,-)$: $W_F \geq 0$ | F11 d=4 group only no theorem from original $J_1, J_2$ | F12 d=8 strict cone $\eta = iJ_2 g$ semigroup branch | F13 d=4 group only $\Theta(+,-)$: $W_F \geq 0$ | F14 d=4 group only no theorem from original $J_1, J_2$ | F15 d=4 strict cone $\eta = J_2 g$ semigroup branch | F16 d=4 group only no theorem from original $J_1, J_2$ |
| -g, I | F17 d=2 group only no theorem from original $J_1, J_2$ | F18 d=2 group only no theorem from original $J_1, J_2$ | F19 d=4 strict cone $\eta = iJ_2 g$ semigroup branch | F20 d=4 group only no theorem from original $J_1, J_2$ | F21 d=4 group only no theorem from original $J_1, J_2$ | F22 d=4 group only no theorem from original $J_1, J_2$ | F23 d=4 group only no theorem from original $J_1, J_2$ | F24 d=2 strict cone $\eta = J_2 g$ semigroup branch |
| -g, -I | F25 d=2 group only $\Theta(+,-)$: $W_F \geq 0$ | F26 d=2 group only no theorem from original $J_1, J_2$ | F27 d=4 strict cone $\eta = iJ_2 g$ semigroup branch | F28 d=4 group only no theorem from original $J_1, J_2$ | F29 d=4 group only no theorem from original $J_1, J_2$ | F30 d=2 group only $\Theta(+,-)$: $W_F \geq 0$ | F31 d=4 group only no theorem from original $J_1, J_2$ | F32 d=4 strict cone $\eta = J_2 g$ semigroup branch |

strict cone: $\eta = J_2 g$ (4/32 per panel)
strict cone: $\eta = iJ_2 g$ (4/32 per panel)
first screen: cone empty by Lemma 1 (16/32)
second screen: remaining cone empty by Lemma 2 (8/32)
outlined group-only cell: anti-linear theorem proves actual weight nonnegative

Primary classification: 8 strict-cone classes and 24 group-only classes per panel. Anti-linear positivity is tested only for the group-only cells: 6 bosonic and 4 fermionic classes have a verified physical representation using their original $J_1, J_2$ pair. Blue bosonic cells use fixed-$N$ weights.